

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	JUNE 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

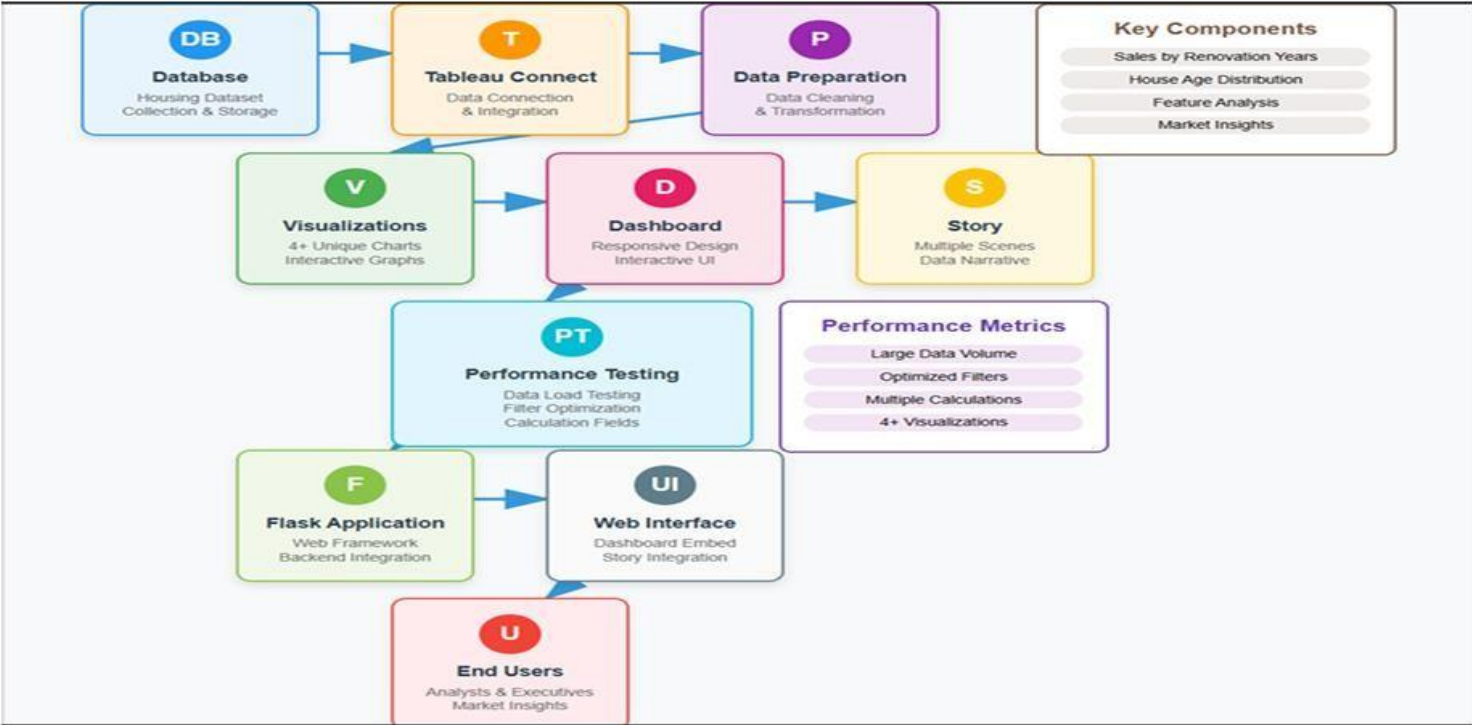


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based dashboards for viewing and interaction	HTML, CSS, JavaScript / Tableau Public
2.	Application Logic-1	Data preprocessing and transformation workflows	Tableau Prep Builder
3.	Application Logic-2	Interactivity using filters, parameters, and actions	Tableau Filters, Parameters, Actions
4.	Dashboard/Story Logic	Logical flow of insights using story features	Tableau Story Feature
5.	Data Source	Flat Files Used as Electricity Consumption Datasets	CSV
6.	File Storage	Heritage datasets stored and accessed locally or from cloud storage	Local File System / Google Drive

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Uses open-source Electricity Consumption datasets, hosted on Tableau Public.	Dataset – Electricity Consumption Patterns (CSV)
2.	Security Implementations	Basic security (view-only links, no edit access on public dashboard)	Tableau Public (limited)
3.	Scalable Architecture	Can scale by publishing to Tableau Cloud or embedding in websites	Tableau Cloud / Web Embed
4.	Availability	Dashboards accessible 24/7 via Tableau Public	Tableau Public
5.	Performance	Performs well with moderate data size; quick filter responsiveness	Tableau Public

References:

<https://c4model.com/> <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
<https://www.ibm.com/cloud/architecture> <https://aws.amazon.com/architecture> <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>